



Xi'an Jiaotong-Liverpool University

西交利物浦大學

FIN 302 Finance and Markets

By Dr. Shan Wu

About Me

- Education background

Exeter University, UK (PhD);



Imperial College London (MSc);



Nanjing Normal University (BEng).



- Before joining XJTLU:

- Nanjing University



- Background:

- Commercial bank work experience

- Research Interests:

- AI and Corporate Governance

Business Sense in important



- Read news everyday
 - Caixin (worth ¥ 900)
<https://libguides.lib.xjtlu.edu.cn/az.php?a=c&p=1>
 - Financial Times
 - Bloomberg
(Trading Floor, IBSS)
 - Reuters
(Refinitive database, Trading Floor, IBSS)
- Read books
- Soft skills

Market Data Company Index Podcast Highlights Events



Weekend Long Read: The Safest and Most Endangered Jobs in a ChatGPT World 🧐

AI will replace simple, repetitive jobs, but those involving complexity, creativity and interpersonal skills will continue to thrive and remain the domain of humans

About this module

- First part
 - Focuses on the financial institutions and the dynamics of financial crises.
- Second part
 - Focuses on the use of financial instruments (e.g., futures, options and swaps as well as credit derivatives) in terms of pricing, trading and risk management (e.g., hedging).
- Textbooks
 - Mishkin, Frederic S., and Stanley G. Eakins, 2018. Ninth Edition. Financial markets and institutions. Pearson Education. ISBN:9781292215006 (For the first part)
 - Hull J.C., 2022. Options Futures and other Derivatives, 11th. Edition, Boston: Pearson. ISBN: 9780136939979 (For the second part)

Assessment

- Group Coursework
 - A recorded video presentation (6-8 minutes).
 - Group size: determined after week 3.
 - Group form by yourselves

Assessment

- Final exam
 - Check exam papers of the **past three years**.
 - FOUR questions. 40% from me and 60% from Prof. He.
 - A formula sheet and tables are included in this examination.
 - A closed-book exam. The university approved calculator can be used.

If you have questions

For FIN302 related questions:

- Ask our TA Yuqi Zhang,
- Ph.D student in Finance
- Yuqi.Zhang2022@student.xjtlu.edu.cn



For general exam-related questions

- IBSS.Assessment@xjtlu.edu.cn

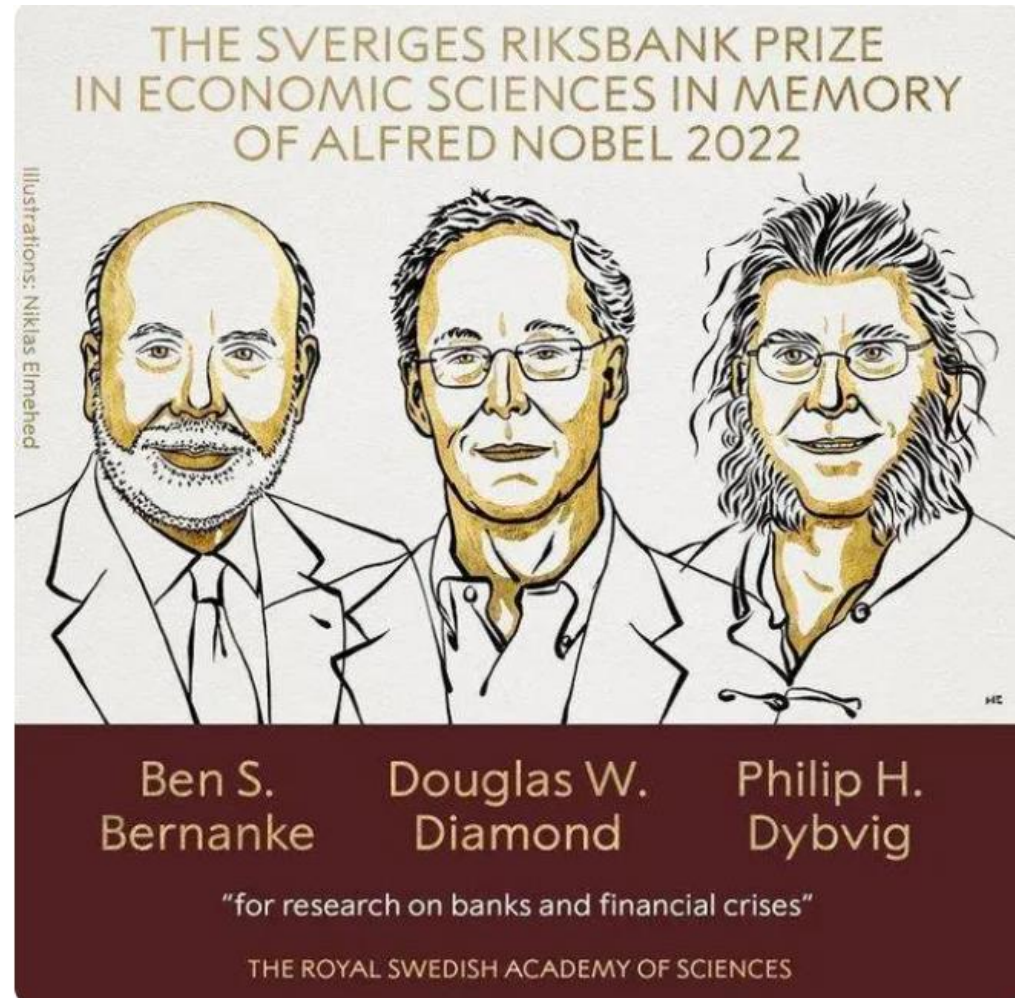
**Please check lecture notes, announcements
before asking questions**

Teaching plan

Week number	Lecture/ Tutorial	Topic	Professor
Week 1	L1	Financial Institutions	Wu
Week 2	L2/T1	Financial Institutions	Wu
Week 3	L3/T2	Financial Institutions& Financial Crisis	Wu
Week 4	L4/T3	Financial Crisis	Wu
Week 5	L5/T4	Financial Crisis	Wu
Week 6	L6/T5	Financial Regulation	Wu
Week 7	L7/T6	Derivatives Markets	He
Week 8	L8/T7	Equity Portfolio Hedging	He
Week 9	L9/T8	Bond Portfolio Hedging	He
Week10	L10/T9	Swaps	He
Week11	L11/T10	Options and Portfolio Insurance	He
Week 12	L12/T11	Option Pricing	He
Week13	L13/T12	Revision	He

Nobel Price in Economic Sciences 2022

- Significantly improved our understanding of **the role of banks** in the economy, particularly during **financial crises**.



This module: Why banks? Why crisis?

Nobel Price in Economic Sciences 2022

1. Why banks? Why don't we lend directly?
 - Delegated Monitoring theory (Diamond)
 - Efficient monitoring
 - Minimizing **informational asymmetries** and **monitoring costs**
2. Why financial crisis? How to mitigate it?
 - Great practical importance in **regulating** financial markets (Bernanke, Diamond and Dybvig)
3. Why liquidity risk is so important?
 - Liquidity risk: you cannot quickly sell assets easily at fair prices
 - **Bank runs** (Diamond).

Nobel Price in Economic Sciences 2022

1.



d)

ries and monitoring

2.

g financial markets

3. Why liquidity risk is so important?

- Liquidity risk: you cannot quickly sell assets easily at fair prices
- **Bank runs** (Diamond)



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FIN 302 Finance and Markets

Why Financial Institutions?

By Dr. Shan Wu

Ref: Chapter 7, Financial markets and institutions. Mishkin

Question:

If you're lack of money...

- If you want to open a company or expand your company, where do you get the money?
 - 3"F": Family, Friend and...
- Financial Institutions (FI).
 - Such as banks, securities, and mutual funds.
- To move funds from people who save, to people who have **productive** investment opportunities
 - We need FI.

Question:

If you have spare money...

- If you have 1 million of retained earnings of your own company, what will you deal with the money?
 - Looking for best investment opportunities.
 - But How?
 - Financial Institutions.
- **A good economy requires a financial system that moves funds from savers to borrowers that most productive.**
- Do you think FIs can do so?
 - Why and why not?

We're going to learn...

Why Financial institutions?

- Basic Facts About Financial Structure
- Transaction Costs
- Asymmetric Information:
 - Adverse Selection
 - Moral Hazard

We're going to learn...

- The Lemons Problem:
 - How **Adverse Selection** Influences Financial Structure
- How **Moral Hazard** Affects the Choice Between Debt and Equity Contracts
- How Moral Hazard Influences Financial Structure in Debt Markets
- Conflicts of Interest

These are prerequisite to understand the financial crises

We're going to learn...

Why Financial institutions?

- Basic Facts About Financial Structure
- Transaction Costs
- Asymmetric Information:
 - Adverse Selection
 - Moral Hazard

8 Facts About Financial Structure Throughout the World

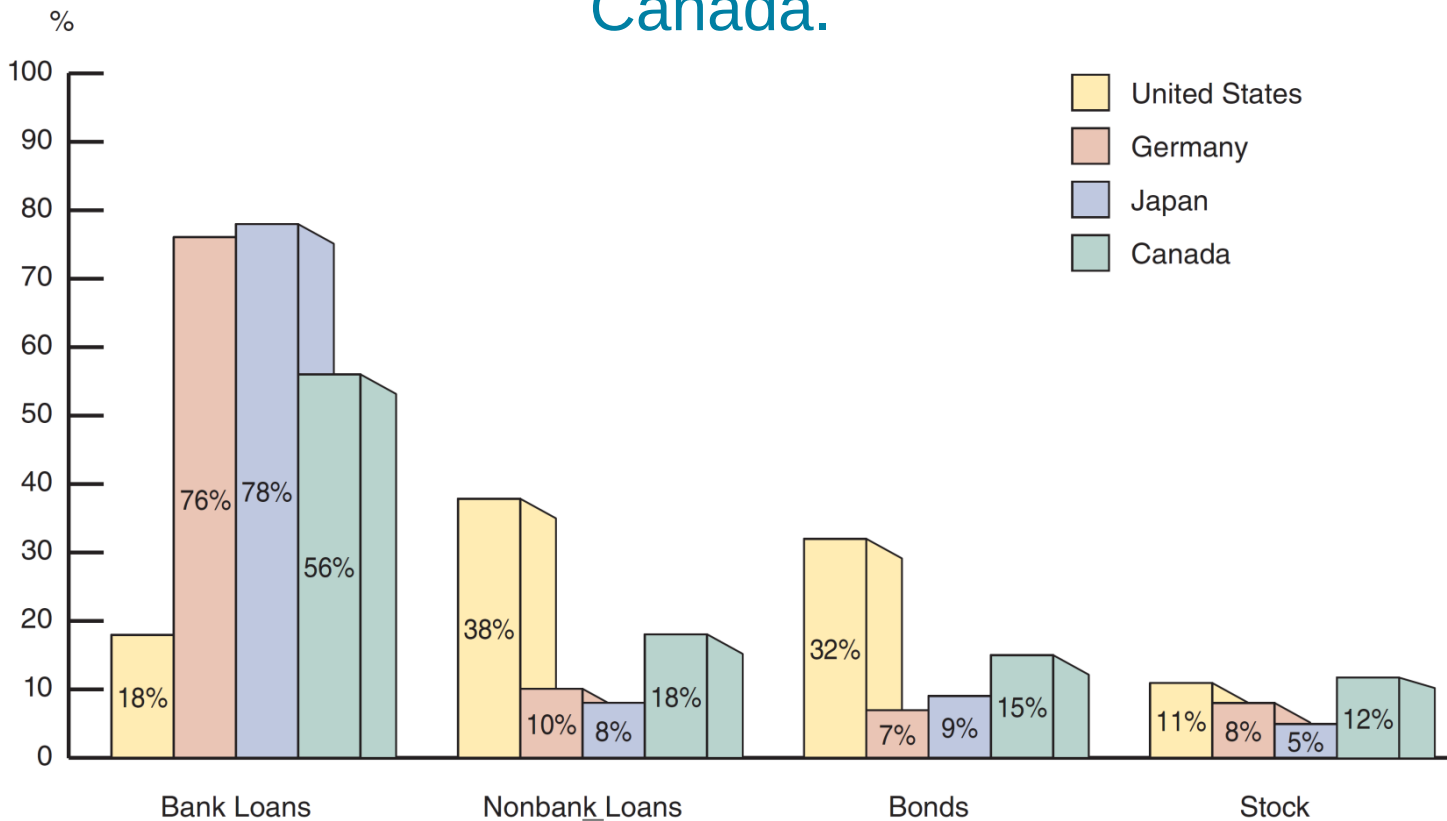
- The financial system
 - A complex structure including many different financial institutions: banks, insurance companies, mutual funds, stock and bonds markets, etc.
- **What is the financial structure of firms across world?**
- Financial structure
 - How firms finance their activities (raise money).
 - E.g. Use banks or not? Borrowing or issue equity?

Source of financing

1. Which one is the most important source of external financing for businesses?
 - Issuing stock
 - Issuing bonds
 - Applying for bank loans

2. Which type is more important for the financing of firms?
 - Financial intermediaries (bank, non-bank)
 - Securities markets (bond, stock, derivatives markets)

**Figure 1 Sources of External Funds for Nonfinancial Businesses:
A Comparison of the United States with Germany, Japan, and
Canada.**

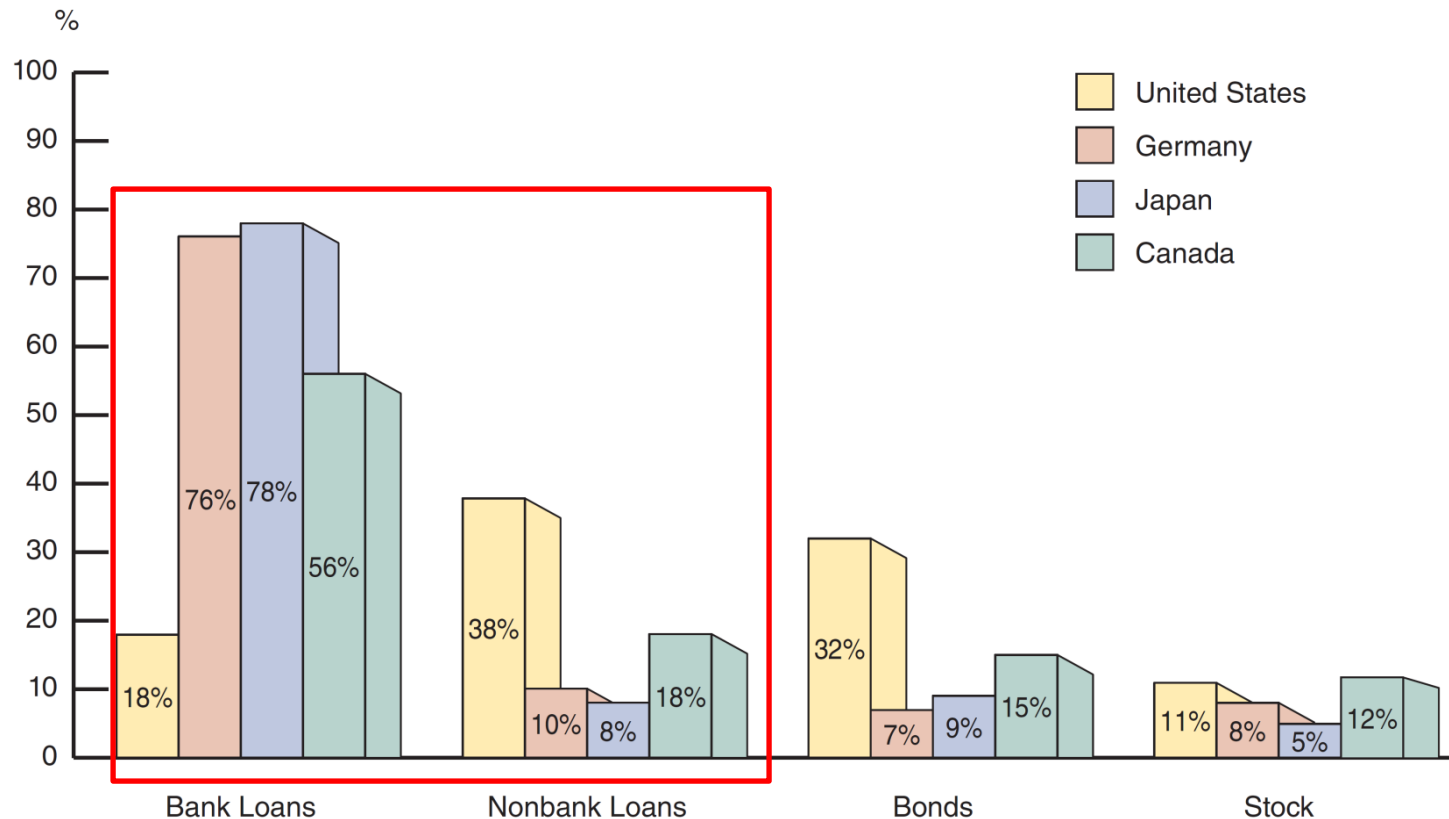


Loans issued by Investment banks, mortgage lenders, money market funds, insurance companies, hedge funds, private equity funds, P2P lenders, trust, etc.

Sources: Andreas Hackethal and Reinhard H. Schmidt, "Financing Patterns: Measurement Concepts and Empirical Results," Johann Wolfgang Goethe-Universitat Working Paper No. 125, January 2004. The data are from 1970–2000 and are gross flows as % of the total, not including trade and other credit data, which are not available.

Facts of Financial Structure

1. Which one is the most important source of external financing for businesses?
2. Which type is more important for the financing of firms?
Financial intermediaries (bank, non-bank) vs. Securities markets (bond, stock, derivatives markets)



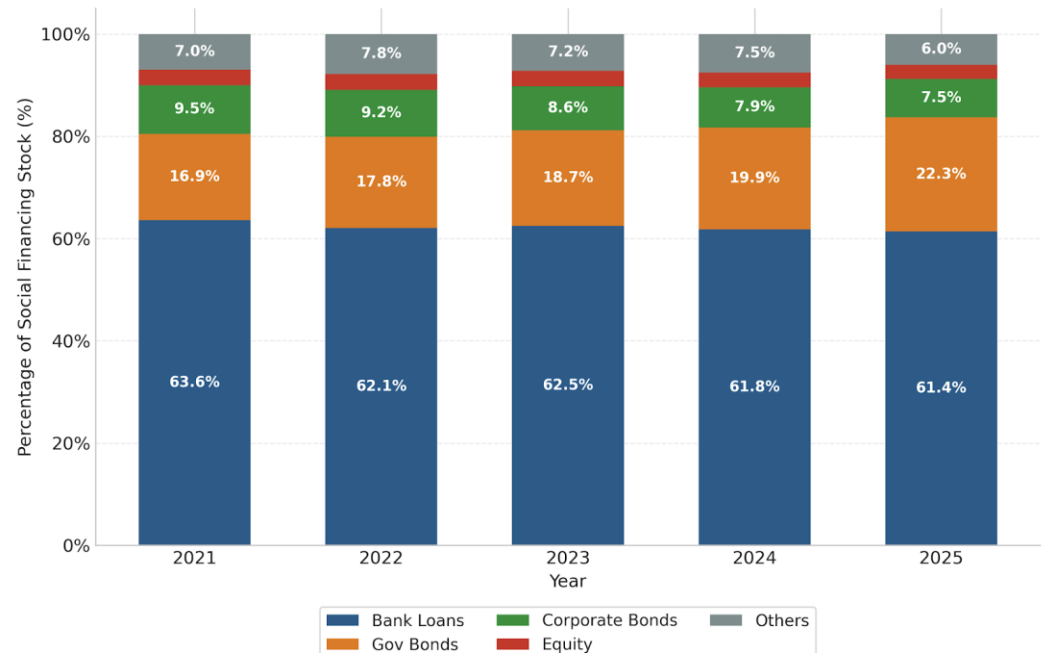
Who is the major sources of external financing?

- Banks have dominated, providing the vast majority of external financing to businesses in China.
 - ~ 70-80% in terms of the total monetary value
- The stock market is estimated to account for around 10-20% of total external financing.
 - Such as IPOs and secondary market transactions
- Hence, **banks are important in the financial market**

8 Facts of Financial Structure

1. Stocks are NOT the most important source of external financing for businesses.
2. Issuing marketable debt and equity securities is NOT the primary way in which businesses finance their operations.
3. Financial intermediaries, particularly **banks**, are the most important source of external funds used to finance businesses.

Financing Structure of Chinese Enterprises (2021-2025)



Facts of Financial Structure

4. Indirect finance vs. Direct finance

- Direct finance
 - Borrowing directly from lenders in the financial market (e.g. issuing stocks; bonds).
 - Eg. Buying Bonds
- Indirect finance
 - Financial intermediary (bank) borrows from you and lends to others.
 - Eg. Bank Deposit
- A metaphor: Buy vegetables from farmers v.s. from the grocery market

Indirect finance vs. Direct finance

- Key difference
- "middleman" (financial institution) buy them first
- Q: Direct finance or Indirect finance?
 1. Buy government bonds;
 2. Apple sells bonds to investors in the open market
- Q: Which is more common to raise funds?
 - Why?
 - Reduced risk? More liquidity? More choice?

Facts of Financial Structure

4. Indirect finance is many times more important than direct finance

- <10% of the external funding is from direct finance
- Why? From borrowers perspective:
- **Low transaction costs**
 - Direct finance is costly, involving issuing securities, underwriting fees, legal compliance, and finding willing investors.
 - Banks have economies of scale in processing loans
- **Providing Liquidity & Risk Transformation**
 - A company may not find immediate buyers for its bonds, but a bank can offer a loan instantly.
 - Banks transform risk by pooling savings and diversifying lending across multiple borrowers.

Facts of Financial Structure

4. Indirect finance is many times more important than direct finance

- Why?
- **Solving Information Asymmetry Problems**
 - Investors lack the expertise or information to assess the creditworthiness of firms
 - Financial intermediaries conduct due diligence, credit scoring, and risk assessment, reducing adverse selection and moral hazard risks.
- **Access for Smaller Firms**
 - Most firms cannot issue bonds or IPOs due to high regulatory costs and lack of investor interest.

Facts of Financial Structure

4. Indirect finance is many times more important than direct finance

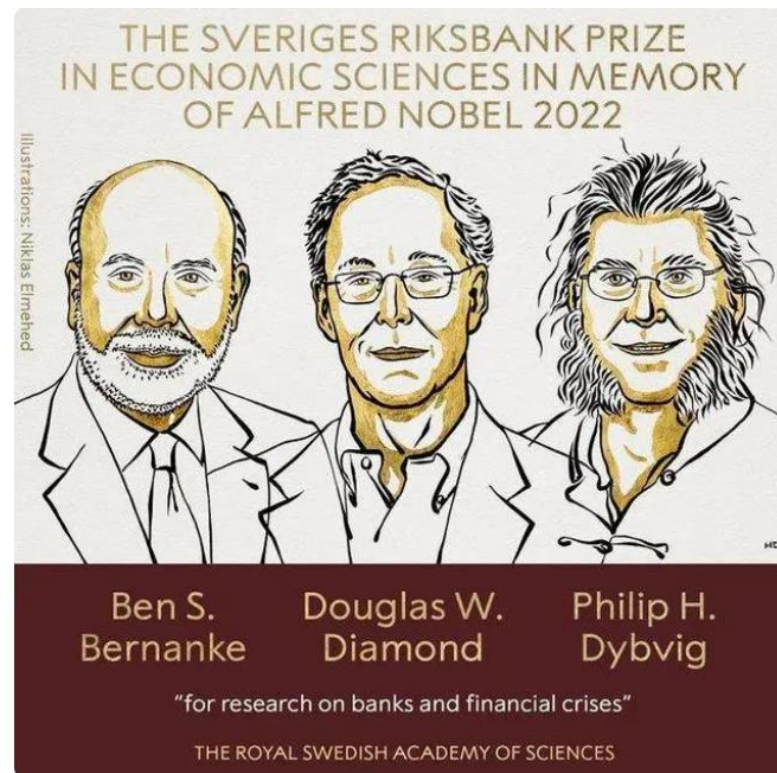
- Why?
- **Market Imperfections & Regulatory Constraints**
 - Stock and bond markets require robust infrastructure, transparency, and investor protection—many economies lack this.
 - Governments often regulate banks to ensure economic stability.
- A declining trend for indirect finance in recent year.
 - Mature in the financial market
 - Fintech and digital lending
 - ...

4/8 facts about financial structure

- 1. Stocks are NOT the most important source of external financing.
- 2. Marketable securities are NOT the primary source of finance.
- 3. Indirect finance is more important than direct finance.
- 4. **Banks are the most important source of external funds.**

Facts of Financial Structure

5. The financial system is among the most **heavily regulated** sectors of economy.
- These 3 scholars showed the importance of regulating financial industry.
 - Why?
 - Highly profitable → High risk
 - → Financial crisis



Facts of Financial Structure

6. Only large, well-established corporations have easy access to securities markets to finance their activities.

- Securities markets
- Markets in stocks, bonds and other securities such as derivatives.
- Why?
 - Cost affordable?
 - Enough investors?

Facts of Financial Structure

6. Only large, well-established corporations have easy access to securities markets to finance their activities.

- Why?
- **High cost**
 - IPO or issuing bonds involves substantial costs
 - *E.g.* For a small company raising \$10 million, fee $\approx$ \$1 million—often unaffordable (7-10% of total proceeds)
- **Information quality $\rightarrow$ Investors' Confidence**
 - Small firms : lack of financial history, limited disclosures, and higher uncertainty.
 - *E.g.* Apple, Microsoft, and Tesla can raise billions in minutes, but a startup struggles to find investors.

Facts of Financial Structure

6. Only large, well-established corporations have easy access to securities markets to finance their activities.

- Why?
- **Liquidity & Market Demand**
 - Securities markets function efficiently when there's high trading liquidity.
 - E.g. Alibaba trade millions of shares daily, whereas small-cap stocks can go days with minimal trades.
- **Market Efficiency & Brand Recognition**
 - Large firms have media coverage, analyst reports, and strong investor relations → transparent and predictable.
 - Investors prefer firms they understand and trust.
 - E.g. Amazon's stock is covered by hundreds of analysts.
 - A small tech startup have no coverage.

Facts of Financial Structure

6. Only large, well-established corporations have easy access to securities markets to finance their activities.

- Why?
- **Alternative Financing for Small Businesses**
 - Bank Loans (Indirect Finance)
 - Venture Capital & Private Equity
 - Angel Investors & Crowdfunding
 - E.g. Sequoia Capital (VC firm)
- Public securities markets favor large, well-established firms due to high fixed costs, investor confidence, liquidity preferences, and regulatory barriers.

7. Collateral is a prevalent feature of debt contracts for both households and businesses.

- What is a Collateral?
 - A property that is pledged to a lender to guarantee payment in the event that the borrower is unable to make debt payments.
- What are the collaterals of these collateralized loans?
 - Car loan
 - Mortgage loan
- Any examples of borrowing without collateral?
 - Credit card; Hua Bei花呗
 - Wangshang Dai蚂蚁金服 “网商贷”

7. Collateral is a prevalent feature of debt contracts for both households and businesses.
- $\frac{1}{4}$ of borrowing by nonfinancial firms is **collateralized debt** (also called secured debt)
 - Why need collateral?
 - Reduce debt default rate

Facts of Financial Structure

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Use Pursuant to Company Instructions

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Summary: 8 facts about financial structure

1. Stocks are not the most important source of external financing
2. Marketable securities are not the primary source of financing
3. **Indirect finance is more important than direct finance**
4. **Banks are the most important source of external funds**
5. The financial system is heavily regulated
6. Only large, well-established firms have access to securities markets
7. Collateral is prevalent in debt contracts
8. Debt contracts have numerous restrictive covenants

We're going to learn...

Why Financial institutions?

- Basic Facts About Financial Structure
- Transaction Costs
- Asymmetric Information:
 - Adverse Selection
 - Moral Hazard

Transaction Costs

- Why financial structure exhibits the 8 facts?
 - Why do borrowers choose this way instead of another way to raise fund?
 - **Cost**
 - Why do lenders choose to put money here instead of there?
 - **Risk and return**
 - Available **information** and **monitoring**
- We need to understand how transaction and information costs affect financial markets
- What is the impact of transaction costs on the structure of our financial system?

Transaction Costs

- What is Transaction costs ?
 - the costs involved in market exchange. These include the costs of discovering market prices and the costs of writing and enforcing contracts.
 - Explicit costs E.g. investment banking fees, legal and accounting expenses, underwriter fees, Brokers' commission fee.
 - Implicit costs E.g. Research time, Regulatory compliance
- Q: What're the transaction costs if you start trading stocks?
 - Brokerage Commission: 0.01% to 0.03% of the transaction value. minimum charge (e.g., ¥5 per trade).
 - Taxes and Regulatory Fees: Stamp Duty, 0.05% (charged to the seller).

Transaction Costs

- Small capital amounts → Higher transaction costs
- Why?
 - Fixed Minimum Commissions
 - Large institutional investors often negotiate "discounts" commissions
 - Costly to search for good stocks
 - Inability to Diversify Efficiently
 - Eg. small trades for a diversified portfolio → Costly
 - The higher the trading frequency, the larger the transaction cost.
 - If large number of small transactions → High transaction costs.

Transaction Costs

- The cost of arranging a purchase of 10,000 shares of stock is not much greater than the cost of arranging a purchase of 50 shares of stock.
 - Transaction cost per trading is relatively high if your purchase is small.
- Due to money constraints, you do not have many choices, cannot diversify your portfolio.



Maotai
¥ 1860/share



Berkshire Hathaway
\$420,000/share

Financial intermediaries reduce Transaction Costs

- Q: Why do we observe more indirect finance than direct finance (fact 3)?
- Financial intermediaries (e.g. mutual funds) make profits by reducing transactions costs for investors
 - Take advantage of economies of scale.
 - Bundling investors' funds together → Reduces transaction costs per individual investors.
 - Develop expertise to lower transactions costs.

Summary

Transaction Costs

- Transaction costs stop small savers and borrowers from direct financial investment.
- Financial intermediaries can help
 - economies of scale
 - develop expertise
- → Reduce transaction costs.